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ARTICLE TYPE

Assignment 2: Data and methodology

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Abstract

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Keywords: keyword entry 1, keyword entry 2, keyword entry 3

1. Introduction & Related Work

This demo file is intended to serve as a “starter file”. It is for preparing manuscript submission only, not for preparing camera-ready versions of manuscripts. Manuscripts will be typeset for publication by the journal, after they have been accepted.

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2. Data Preparation & Pipeline Design

The dataset is taken from the internal `Staging` database at my organization (our internal sales calls using this system). I selected rows with **SQL** and exported the result to CSV.

```
SELECT *
FROM call_call
WHERE type='MANUAL'
  AND record != ''
  AND duration > 5
ORDER BY created_at DESC;
```

`duration > 5` excludes empty or very short calls. The export contains **13,232 records.**

2.1 Data Visualization

Figure 1 plots `duration_sec` for all **13,232** calls with valid duration. The distribution is right-skewed: **most calls are short.** Durations above the 99th percentile (about **775 s**) are clipped and grouped in the final bin so the main mass of the histogram remains visible.

In a call, the silence is often much longer than the speech. This **motivates segmenting audio** before speech-to-text instead of processing entire files end-to-end.

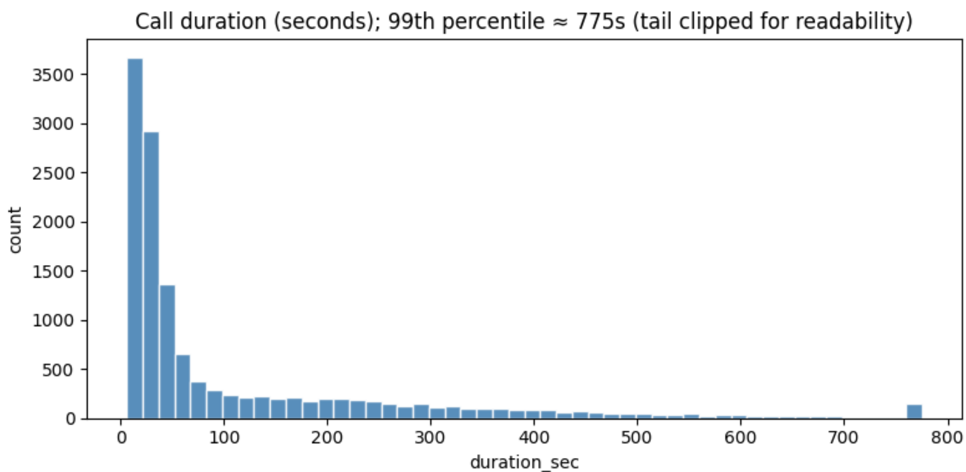

Figure 1. Distribution of call duration (seconds). Values above the 99th percentile are merged into the last bin.

2.2 Why do we need to preprocess the audio?

The goal is to feed a speech-to-text model **audio that is mostly speech.** Raw files add silence, hold tones, background noise, and non-speech events. That increases recognition errors. Preprocessing limits those segments before ASR.

Recordings are stereo: agent on the left channel and customer on the right in my setup. Downstream analysis targets the customer, so I take the right channel before VAD.

Voice activity detection (VAD) outputs time intervals with speech. I retain those intervals and drop the rest. That removes long silent regions and much background sound prior to ASR. It is an imperfect but standard front end.

2.3 Pre-processing Step 1: Voice Activity Detection (VAD)

I use **Silero VAD** (Silero Team 2024): low latency on CPU/GPU, suitable for batch jobs, and fast enough for real-time use.

Figure 2 is one example from my Silero VAD test. Left: both channels overlaid (labels as in my notebook). Right: customer channel only, with green spans for VAD-detected speech. Audio is resampled to **16 kHz** for the model.

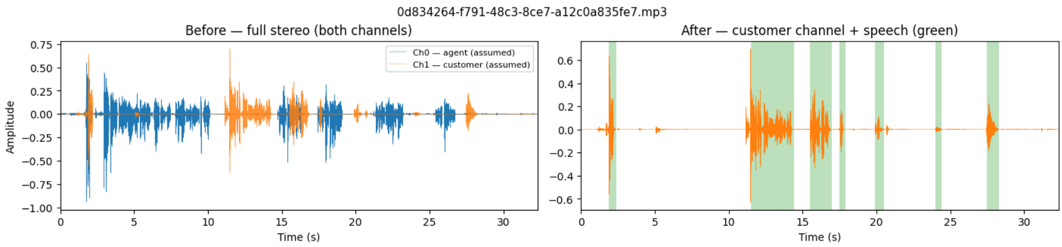

Figure 2. One call: stereo mix (left) and customer channel with VAD speech regions (right).

2.4 Pre-processing Step 2: Automatic Speech Recognition (ASR)

After VAD segmentation, each speech-only chunk is forwarded to an Automatic Speech Recognition (ASR) model. In this project, I use the Vietnamese wav2vec2 checkpoint `nguyenvulebinh/wav2vec2-base-vi` Binh 2022. It follows the wav2vec2 architecture and is self-supervised pre-trained on Vietnamese speech. The ASR model converts audio segments into unstructured text transcripts. Segment-level transcripts are then merged by timestamp to reconstruct each full-call transcript.

2.5 Step 3: LLM Fine-Tuning

In the final stage, reconstructed transcripts are turned into a supervised fine-tuning dataset for the downstream LLM. Each sample is a JSON record with fields `instruction`, `input`, and `response`. The `instruction` should spell out role, task, constraints, and the exact output contract (so the model sees a stable prompt at train and inference time); `input` is the full transcript; `response` is the target value only.

Two training rows for the same transcript (one per field) are shown below; `response` types match the two-column field reference in this subsection.

```
{
  "instruction": "You are a structured-field extractor for telesales QA. The user message
  ↪   contains one full call transcript (agent and customer).\n\nTask: assign
  ↪   customer_need_category.\n\nRules:\n- Return exactly one snake_case label from the closed
  ↪   set listed under that field in the schema (see the two-column reference).\n- Rely only
  ↪   on what the customer says or clearly implies; do not invent facts.\n- If the transcript
  ↪   does not support any label, return no_information.\n- Output the label token only: no
  ↪   quotes, commas, JSON, or explanation.",
  "input": "<transcript>",
  "response": "needs_more_information"
}
{
  "instruction": "You are a structured-field extractor for telesales QA. The user message
  ↪   contains one full call transcript (agent and customer).\n\nTask: extract
  ↪   customer_callback_time.\n\nRules:\n- If the customer agrees to a callback and a calendar
  ↪   date is stated or unambiguously implied, return it as YYYY-MM-DD in Gregorian
  ↪   calendar.\n- If no callback is agreed or the date cannot be resolved, return JSON null
  ↪   (bare null, not a string).\n- Output only the date string or null, with no surrounding
  ↪   text.",
  "input": "<transcript>",
  "response": "2026-04-06"
}
```

This schema fixes supervision targets so the model can be fine-tuned on consistent instruction–response pairs.

Example fields

```
column: customer_need_category
type: classification
groups:
  interested | not_interested
  | no_information | needs_more_information
  | price_or_budget_concern | timing_defer
  | comparing_options | already_satisfied
  | undecided
response: Enum[str]
```

```
column: customer_callback_time
type: optional_date
response: date | null
format: YYYY-MM-DD
notes:
  bare JSON null if unknown
```

3. Deep Learning Methodology

The output for a single-column figure is in Figure 3. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

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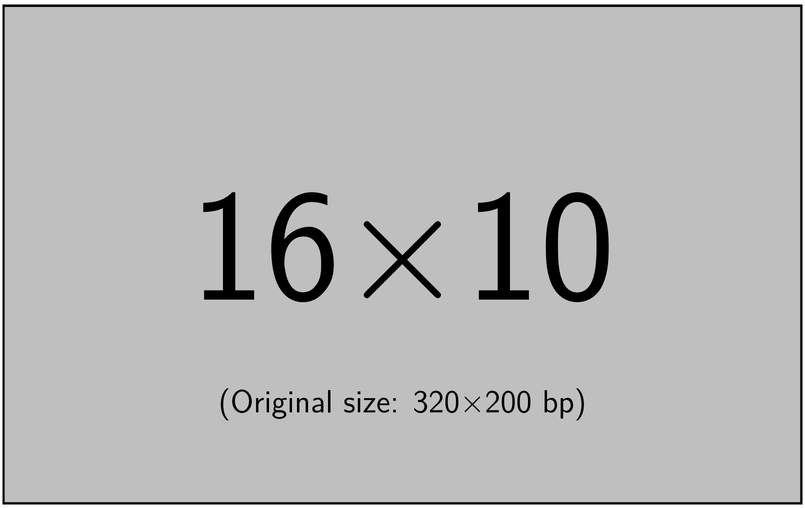

Figure 3. Insert figure caption here

See example table in Table 1.

Table 1. An Example of a Table

Column head 1	Column head 2	Column head 3	Column head 4
One[a]	Two	three three	four
Three	Four	three three[b]	four

Table note
a First note
b Another table note

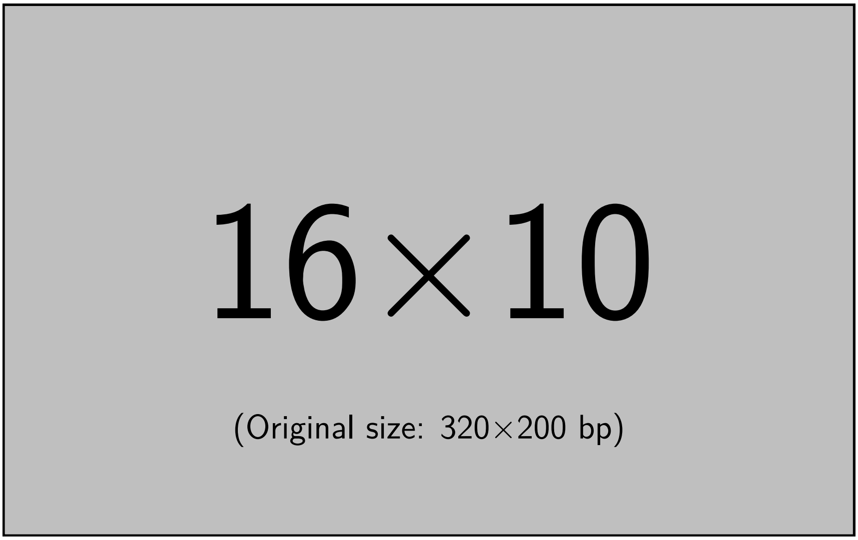

Figure 4. Insert figure caption here

4. Evaluation Metrics

The conclusion text goes here.

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Competing Interests A statement about any financial, professional, contractual or personal relationships or situations that could be perceived to impact the presentation of the work — or 'None' if none exist

Data Availability Statement A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings — e.g. Replication data and code can be found in Harvard Dataverse: `\url{https://doi.org/link}`.

Ethical Standards The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions Please provide an author contributions statement using the CRediT taxonomy roles as a guide `\url{https://www.casrai.org/credit.html}`. Conceptualization: A.A; A.B. Methodology: A.A; A.B. Data curation: A.C. Data visualisation: A.C. Writing original draft: A.A; A.B. All authors approved the final submitted draft.

Notes

1 A footnote/endnote

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Appendix 1. Example Appendix Section

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